Brutal Practice Paper B15 - Winds, Storms & Cyclones / Integers / Motion & Time / Algebraic Expressions

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. A cyclone's winds blow at 180 km/h. Express this speed in metres per second. [Winds, Storms & Cyclones]

- **(A)** 100 m/s
- **(B)** 60 m/s
- **(C)** 5 m/s
- **(D)** 36 m/s
- **(E)** 50 m/s
- **(F)** 648 m/s

2. Evaluate: (-15) + 24 + (-9) + (-6). [Integers]

- **(A)** 6
- **(B)** 0
- **(C)** -12
- **(D)** -6
- **(E)** -36
- **(F)** 12
- **(G)** -24

3. A train covers 240 km in 3 hours. Find its average speed in km/h, and then in m/s. [Motion & Time]

- **(A)** 720 km/h, which is 200 m/s
- **(B)** 80 km/h, which is 8 m/s
- **(C)** 80 km/h, which is about 288 m/s
- **(D)** 80 km/h, which is about 22.2 m/s
- **(E)** 60 km/h, which is 16.7 m/s
- **(F)** 80 km/h, which is 80 m/s
- **(G)** 0.0125 km/h, which is tiny

4. In the expression 7 - 5x + 2x squared, what is the coefficient of x? [Algebraic Expressions]

- **(A)** 1
- **(B)** -7
- **(C)** 2
- **(D)** -5
- **(E)** -2
- **(F)** 7
- **(G)** 5

5. Two weather stations read the air pressure: station A shows 1006 hPa and station B shows 994 hPa. Surface wind blows from higher to lower pressure. What is the pressure difference and which way will the wind blow? [Winds, Storms & Cyclones]

- **(A)** 12 hPa, from A to B
- **(B)** 2000 hPa, from A to B
- **(C)** 12 hPa, from B to A
- **(D)** 6 hPa, from A to B
- **(E)** 12 hPa, so no wind blows
- **(F)** 1000 hPa, from A to B
- **(G)** 18 hPa, from A to B

6. A diver goes down to -45 m, then rises 18 m, then descends another 27 m. At what depth is the diver now? [Integers]

- **(A)** -90 m
- **(B)** -54 m
- **(C)** 0 m
- **(D)** 36 m
- **(E)** -36 m
- **(F)** -18 m

7. A car travels 60 km at 60 km/h, then another 60 km at 30 km/h. What is its average speed for the whole journey? [Motion & Time]

- **(A)** 45 km/h
- **(B)** 90 km/h
- **(C)** 50 km/h
- **(D)** 40 km/h
- **(E)** 60 km/h
- **(F)** 120 km/h
- **(G)** 30 km/h

8. Find the value of 3x squared - 2x + 5 when x = -2. [Algebraic Expressions]

- **(A)** 25
- **(B)** 9
- **(C)** 21
- **(D)** 13
- **(E)** 5
- **(F)** 19

9. While playing, you blow a fast stream of air between two light paper strips hanging side by side. What do the strips do, and why? [Winds, Storms & Cyclones]

- **(A)** They flap up and down only
- **(B)** They move towards each other, because fast air has lower pressure
- **(C)** They move apart, because fast air has higher pressure
- **(D)** They get heavier and drop
- **(E)** They first touch then fly apart
- **(F)** They stay perfectly still
- **(G)** They spin in a circle

10. Evaluate: (-8) x (-7) + (-6) x 5. [Integers]

(A) -56
(B) 2
(C) 26
(D) -26
(E) 86
(F) -86
(G) -2

11. A pendulum makes 30 complete oscillations in 60 seconds. What is its time period? [Motion & Time]

(A) 0.5 s
(B) 1800 s
(C) 30 s
(D) 60 s
(E) 2 s
(F) 4 s
(G) 0.2 s

12. Add the expressions (4a - 3b + 2) and (-a + 5b - 7). [Algebraic Expressions]

(A) 3a + 8b - 5
(B) 3a + 2b + 9
(C) 3a + 2b - 5
(D) 3a - 2b - 5
(E) -3a + 2b - 5
(F) 5a + 2b - 5
(G) 3a + 2b - 9

13. A cyclone is 260 km out at sea and moving straight towards the coast at a steady 20 km/h. If the time now is 06:00, when will it make landfall? [Winds, Storms & Cyclones]

(A) 07:00 the same day
(B) next year
(C) 13:00 the same day
(D) 19:00 the same day
(E) 12:30 the same day
(F) 18:00 the same day
(G) 08:00 next day

14. Evaluate: (-72) / 9 - (-4) x 3. [Integers]

(A) 20
(B) -12
(C) 12
(D) -4
(E) -20
(F) 4

15. Convert a speed of 90 km/h into metres per second. [Motion & Time]

(A) 16 m/s
(B) 250 m/s
(C) 2.5 m/s
(D) 50 m/s
(E) 25 m/s
(F) 324 m/s
(G) 18 m/s

16. Subtract (2x - 3y) from (5x + y). [Algebraic Expressions]

(A) 3x + 4y
(B) 7x - 2y
(C) 3x - 4y
(D) 7x + 4y
(E) -3x - 4y
(F) 3x - 2y

17. During an approaching storm the barometer falls a steady 4 hPa every hour. It now reads 1008 hPa. What will it read 5 hours later, and what is the total fall? [Winds, Storms & Cyclones]

(A) 988 hPa, a fall of 40 hPa
(B) 988 hPa, a fall of 20 hPa
(C) 988 hPa, a fall of 4 hPa
(D) 968 hPa, a fall of 40 hPa
(E) 1004 hPa, a fall of 4 hPa
(F) 990 hPa, a fall of 18 hPa

18. The temperature is 5 deg C and then falls a steady 3 deg C each hour for 4 hours. What is the new temperature? [Integers]

(A) -17 deg C
(B) -3 deg C
(C) 17 deg C
(D) 2 deg C
(E) -12 deg C
(F) -7 deg C
(G) 7 deg C

19. Convert a speed of 15 m/s into kilometres per hour. [Motion & Time]

(A) 150 km/h
(B) 4.17 km/h
(C) 15 km/h
(D) 75 km/h
(E) 60 km/h
(F) 54 km/h
(G) 27 km/h

20. How many terms are there in the expression 5xy - 3x + 7y squared - 2? [Algebraic Expressions]

- **(A)** 7
- **(B)** 3
- **(C)** 6
- **(D)** 2
- **(E)** 4
- **(F)** 5
- **(G)** 1

21. What is the calm, nearly cloud-free centre of a cyclone called? [Winds, Storms & Cyclones]

- **(A)** the tail
- **(B)** the trough
- **(C)** a spiral band
- **(D)** the eye
- **(E)** the front
- **(F)** the storm surge
- **(G)** the eye-wall

22. Evaluate: (-2) x (-2) x (-2) x (-2). [Integers]

- **(A)** 8
- **(B)** 16
- **(C)** 4
- **(D)** -8
- **(E)** 32
- **(F)** -16
- **(G)** -32

23. A bus leaves at 08:45 and arrives at 11:15, covering 150 km. What was its average speed? [Motion & Time]

- **(A)** 50 km/h
- **(B)** 100 km/h
- **(C)** 375 km/h
- **(D)** 60 km/h
- **(E)** 75 km/h
- **(F)** 2.5 km/h
- **(G)** 30 km/h

24. Write an expression for 'five less than three times a number n'. [Algebraic Expressions]

- **(A)** 3(n - 5)
- **(B)** 5 - 3n
- **(C)** 3n - 5
- **(D)** 15n
- **(E)** 3n + 5
- **(F)** 5n - 3
- **(G)** n/3 - 5

25. Over a warm ocean, moist air rises, the water vapour condenses and releases heat, which makes even more air rise. What does this chain mainly explain? [Winds, Storms & Cyclones]

- **(A)** Why a cyclone dies over warm seas
- **(B)** Why cyclones spin slower over the sea
- **(C)** Why a warm ocean stops a cyclone
- **(D)** Why cyclones need cold dry air
- **(E)** Why a tropical cyclone grows stronger over warm seas
- **(F)** Why cyclones form over deserts
- **(G)** Why cyclones form over mountains

26. A quiz awards +5 for a correct answer and -2 for a wrong one. Riya gets 12 correct and 8 wrong. What is her score? [Integers]

- **(A)** 76
- **(B)** 44
- **(C)** 60
- **(D)** 20
- **(E)** 52
- **(F)** 36
- **(G)** -44

27. An odometer reads 45670 km at the start of a drive and 45920 km at the end, 5 hours later. What was the average speed? [Motion & Time]

- **(A)** 50 km/h
- **(B)** 250 km/h
- **(C)** 5 km/h
- **(D)** 60 km/h
- **(E)** 91500 km/h
- **(F)** 9184 km/h

28. Among 3x squared y, -7xy squared, 5yx squared and 2x squared y, which terms are LIKE terms? [Algebraic Expressions]

- **(A)** none are like terms
- **(B)** 3x squared y, 5yx squared and 2x squared y
- **(C)** 3x squared y and 5yx squared only
- **(D)** all four terms
- **(E)** -7xy squared and 2x squared y
- **(F)** 3x squared y and -7xy squared
- **(G)** 3x squared y and 2x squared y only

29. The wind speed at a coast rises from 30 km/h to 120 km/h as a cyclone nears. By what factor did the speed increase, and what generally happens to the air pressure there? [Winds, Storms & Cyclones]

- **(A)** 3 times faster, and the pressure drops
- **(B)** 90 times faster, and the pressure drops
- **(C)** 4 times faster, and the pressure drops
- **(D)** 4 times faster, and the pressure rises
- **(E)** 40 times faster, and the pressure drops
- **(F)** 2 times faster, and the pressure drops
- **(G)** 4 times faster, and the pressure stays the same

30. Evaluate: 18 - (-6) x (-2) + (-4). [Integers]

(A) -2
(B) 26
(C) 10
(D) -14
(E) -26
(F) 2
(G) 14

31. A pendulum has a time period of 1.5 s. How many complete oscillations does it make in 1 minute? [Motion & Time]

(A) 30
(B) 45
(C) 90
(D) 40
(E) 20
(F) 60
(G) 1.5

32. Simplify: 3x + 2y - 5x + 7y. [Algebraic Expressions]

(A) -2x + 9y
(B) -2x + 5y
(C) 8x - 5y
(D) -2x - 9y
(E) -8x + 9y
(F) 2x + 9y
(G) 8x + 9y

33. A storm pushes a 3 m sea surge onto a coast where the tide is already 2 m above normal, and waves add a further 1 m on top. How high does the water rise above normal sea level? [Winds, Storms & Cyclones]

(A) 4 m
(B) 3 m
(C) 1 m
(D) 2 m
(E) 32 m
(F) 6 m

34. The product of two integers is -48. One of them is -6. What is the other integer? [Integers]

(A) 288
(B) 8
(C) -288
(D) -42
(E) 6
(F) 42
(G) -8

35. On a distance-time graph, what does a flat horizontal line mean about the object? [Motion & Time]

- **(A)** It is speeding up
- **(B)** It is falling
- **(C)** It is moving very fast
- **(D)** It is moving at constant speed
- **(E)** It is moving backward
- **(F)** It is slowing down
- **(G)** It is at rest (not moving)

36. Find the value of 2a squared b when a = 3 and b = -1. [Algebraic Expressions]

- **(A)** -18
- **(B)** -6
- **(C)** 18
- **(D)** 6
- **(E)** -36
- **(F)** 12
- **(G)** -12

37. Which order shows the stages of a cyclone reaching a coast from EARLIEST to LAST? [Winds, Storms & Cyclones]

- **(A)** Low pressure forms over warm sea, winds spiral in, eye crosses the coast, surge floods the land
- **(B)** Surge floods, winds spiral in, eye crosses the coast, low pressure forms
- **(C)** Surge floods the land, eye crosses the coast, winds spiral in, low pressure forms
- **(D)** Low pressure forms, surge floods, eye crosses the coast, winds spiral in
- **(E)** Eye crosses the coast, low pressure forms, winds spiral in, surge floods
- **(F)** Winds spiral in, surge floods, low pressure forms, eye crosses the coast
- **(G)** Eye crosses the coast, surge floods, low pressure forms, winds spiral in

38. What is the sum of all the integers from -7 to 7, including both ends? [Integers]

- **(A)** -14
- **(B)** 7
- **(C)** -7
- **(D)** 49
- **(E)** 0
- **(F)** 1

39. A cheetah runs at 30 m/s. How far does it travel in 1 minute, expressed in kilometres? [Motion & Time]

- **(A)** 0.5 km
- **(B)** 0.03 km
- **(C)** 3.6 km
- **(D)** 180 km
- **(E)** 1.8 km
- **(F)** 1800 km

40. What is the degree of the expression 4x cubed - 2x squared y squared + 7? [Algebraic Expressions]

- **(A)** 4
- **(B)** 2
- **(C)** 6
- **(D)** 7
- **(E)** 5
- **(F)** 3
- **(G)** 1

41. A village 48 km from the coast must clear out before a cyclone that will arrive in 3 hours. Evacuation buses travel at 40 km/h. Can a bus reach the coast-road safe zone in time, and how long is one trip? [Winds, Storms & Cyclones]

- **(A)** Yes, because one trip takes 1.2 hours, which is under 3 hours
- **(B)** Yes, because one trip takes 12 hours
- **(C)** No, because one trip takes 3 hours
- **(D)** Yes, because one trip takes 0.83 hours
- **(E)** Yes, because one trip takes 2 hours
- **(F)** Yes, because one trip takes 1920 hours
- **(G)** No, because one trip takes 1.2 hours

42. Evaluate: [(-100) / (-10)] / (-5). [Integers]

- **(A)** -2
- **(B)** 50
- **(C)** 10
- **(D)** 2
- **(E)** -20
- **(F)** 20
- **(G)** -50

43. Two stations are 360 km apart. A train leaves at 10:00 travelling at a steady 90 km/h. At what time does it arrive? [Motion & Time]

- **(A)** 13:00
- **(B)** 16:00
- **(C)** 12:00
- **(D)** 14:30
- **(E)** 18:00
- **(F)** 14:00

44. A triangle has sides (2x + 1), (3x - 2) and (x + 4). Write an expression for its perimeter. [Algebraic Expressions]

- **(A)** 6x - 1
- **(B)** 5x + 7
- **(C)** 6x + 7
- **(D)** 6x - 3
- **(E)** 5x + 3
- **(F)** 6x + 3

45. Which instrument is used to measure wind speed? [Winds, Storms & Cyclones]

(A) thermometer
(B) rain gauge
(C) anemometer
(D) hygrometer
(E) barometer
(F) wind vane
(G) seismograph

46. A submarine at -200 m descends in 5 equal steps of -25 m each. What is its final depth? [Integers]

(A) -325 m
(B) -75 m
(C) 325 m
(D) 75 m
(E) -225 m
(F) -300 m

47. On a distance-time graph, what does a steeper (more sloped) straight line mean? [Motion & Time]

(A) The object is stopped
(B) A negative time
(C) A greater speed
(D) The same speed as a gentle line
(E) A lower speed
(F) The object is at rest
(G) The object moves backward

48. Subtract the sum of (3a - 2b) and (a + b) from 10a. [Algebraic Expressions]

(A) -6a + b
(B) 6a + b
(C) 4a - b
(D) 14a - b
(E) 6a - 3b
(F) 6a - b

49. At night a land breeze blows from the land out to the sea. Why? [Winds, Storms & Cyclones]

(A) Land cools faster than the sea, so air over the warmer sea rises and air flows out from land to fill it
(B) Wind always blows towards the sea
(C) The land is always hotter than the sea at night
(D) Sea water boils at night and pushes air
(E) The Moon pulls the air outward
(F) The sea cools faster than the land at night
(G) There is never any breeze at night

50. Evaluate: (-15) x 0 + (-9) / (-9) - 7. [Integers]

- **(A)** -7
- **(B)** 8
- **(C)** 0
- **(D)** -8
- **(E)** -6
- **(F)** 6

51. Sunlight takes 500 s to reach the Earth, travelling at about 300000 km/s. How far is the Sun from the Earth? [Motion & Time]

- **(A)** 3,000,000 km
- **(B)** 150,000 km
- **(C)** 1,500,000,000 km
- **(D)** 150,000,000 km
- **(E)** 600 km
- **(F)** 1,500,000 km
- **(G)** 15,000,000 km

52. Simplify: 5(x - 2) - 3(x - 4). [Algebraic Expressions]

- **(A)** 8x + 2
- **(B)** 8x - 22
- **(C)** 2x - 2
- **(D)** 2x + 22
- **(E)** 2x + 2
- **(F)** 2x - 22
- **(G)** 2x + 14

53. A rain gauge collects 90 mm of rain during a 6-hour storm. What is the average rainfall rate in centimetres per hour? [Winds, Storms & Cyclones]

- **(A)** 9 cm/h
- **(B)** 2.5 cm/h
- **(C)** 540 cm/h
- **(D)** 6 cm/h
- **(E)** 1.5 cm/h
- **(F)** 15 cm/h
- **(G)** 0.15 cm/h

54. Fill the blank: (-6) x ___ = 54. What number goes in the blank? [Integers]

- **(A)** 60
- **(B)** -60
- **(C)** -9
- **(D)** -324
- **(E)** 48
- **(F)** 9

55. A car moves at 72 km/h. How many metres does it cover in just 1 second? [Motion & Time]

(A) 259.2 m
(B) 36 m
(C) 72 m
(D) 200 m
(E) 20 m
(F) 18 m

56. What is the constant term in the expression 9 - 4x + x squared? [Algebraic Expressions]

(A) 1
(B) -9
(C) 9
(D) x squared
(E) 4
(F) 0

57. During a thunderstorm, what is the safest action? [Winds, Storms & Cyclones]

(A) Lie flat in a pool of water
(B) Touch metal railings to feel safe
(C) Stay indoors, away from windows and metal pipes
(D) Hold up an umbrella with a metal rod
(E) Shelter under a tall lone tree
(F) Stand on the top of a hill
(G) Stand in the middle of an open field

58. A lift starts at floor +3, goes down 7 floors, up 12 floors, then down 5 floors. At which floor does it stop? [Integers]

(A) +9
(B) -9
(C) +27
(D) -3
(E) +3
(F) 0
(G) +7

59. A wheel turns 5 complete times every second. How long does one full turn take? [Motion & Time]

(A) 0.02 s
(B) 5 s
(C) 1 s
(D) 2 s
(E) 0.2 s
(F) 10 s
(G) 0.5 s

60. If P = 2x + 3 and Q = x - 5, find P - Q. [Algebraic Expressions]

- **(A)** -x + 8
- **(B)** 3x + 8
- **(C)** x - 2
- **(D)** x + 2
- **(E)** 3x - 2
- **(F)** x - 8
- **(G)** x + 8

61. Three towns record midday pressures: P = 1002 hPa, Q = 998 hPa, R = 1010 hPa. Surface wind blows from the highest towards the lowest pressure. Between which two towns is the strongest pull, and how big is that difference? [Winds, Storms & Cyclones]

- **(A)** From P to Q, a difference of 4 hPa
- **(B)** From R to P, a difference of 8 hPa
- **(C)** From P to R, a difference of 12 hPa
- **(D)** From Q to R, a difference of 12 hPa
- **(E)** From R to Q, a difference of 2010 hPa
- **(F)** From R to Q, a difference of 12 hPa

62. Write -13 + (-13) + (-13) as a multiplication and find its value. [Integers]

- **(A)** 3 x (-13) = -16
- **(B)** 3 x (-13) = -3
- **(C)** 3 x (-13) = -42
- **(D)** 3 x (-13) = -39
- **(E)** 2 x (-13) = -26
- **(F)** 3 x (-13) = 39
- **(G)** (-13) x (-13) = 169

63. A runner does 400 m in 50 s, rests for 10 s, then does another 400 m in 50 s. What is the average speed over the whole 110 s? [Motion & Time]

- **(A)** about 14.5 m/s
- **(B)** 6.7 m/s
- **(C)** 800 m/s
- **(D)** about 7.3 m/s
- **(E)** 16 m/s
- **(F)** 4 m/s
- **(G)** 8 m/s

64. Find the product of 3x and -4x squared. [Algebraic Expressions]

- **(A)** -12x cubed
- **(B)** 12x cubed
- **(C)** -12x
- **(D)** 12x squared
- **(E)** -x cubed
- **(F)** -7x cubed
- **(G)** -12x squared

65. Tropical cyclones in the Bay of Bengal mainly form over which kind of surface? [Winds, Storms & Cyclones]

- **(A)** dry river beds
- **(B)** thick forests
- **(C)** cold polar seas
- **(D)** ice sheets
- **(E)** high mountain peaks
- **(F)** warm ocean water

66. Which is the GREATEST of these integers: -2, -100, 0, -7, -1? [Integers]

- **(A)** none of these
- **(B)** 0
- **(C)** -7
- **(D)** -1
- **(E)** -100
- **(F)** -2
- **(G)** -101

67. What is the basic (SI) unit of time? [Motion & Time]

- **(A)** oscillation
- **(B)** metre
- **(C)** hour
- **(D)** minute
- **(E)** year
- **(F)** second
- **(G)** day

68. A rectangle is (3x) long and (2x - 1) wide. Write an expression for its perimeter. [Algebraic Expressions]

- **(A)** 10x - 4
- **(B)** 10x - 2
- **(C)** 6x - 2
- **(D)** 10x + 2
- **(E)** 10x - 1
- **(F)** 6x squared - 3x

69. In a thunderstorm you see a lightning flash and hear the thunder 6 seconds later. Sound travels about 330 m/s and light reaches you almost at once. About how far away is the lightning? [Winds, Storms & Cyclones]

- **(A)** about 55 m
- **(B)** about 198 m
- **(C)** about 1980 m
- **(D)** about 330 m
- **(E)** about 660 m
- **(F)** about 3960 m

70. Evaluate: (56 / -8) x (-3) + (-10). [Integers]

- **(A)** -1
- **(B)** 21
- **(C)** 1
- **(D)** -31
- **(E)** -11
- **(F)** 11
- **(G)** 31

71. If a simple pendulum is made longer, what happens to its time period? [Motion & Time]

- **(A)** It stays exactly the same
- **(B)** It increases
- **(C)** It becomes negative
- **(D)** It depends on the bob's colour
- **(E)** It decreases
- **(F)** It doubles the speed
- **(G)** It becomes zero

72. Find the value of x squared - y squared when x = 7 and y = 3. [Algebraic Expressions]

- **(A)** 4
- **(B)** 58
- **(C)** 16
- **(D)** 20
- **(E)** 100
- **(F)** 40
- **(G)** -40

73. Fast wind blows over a flat roof while the air pressure beneath the roof stays normal. What happens to the roof? [Winds, Storms & Cyclones]

- **(A)** It is pushed upward and can be torn off
- **(B)** It is pressed down more firmly
- **(C)** It becomes much colder
- **(D)** Nothing happens to the roof
- **(E)** It only slides sideways
- **(F)** Pressure stays equal so it is safe
- **(G)** It becomes heavier

74. A bank account is overdrawn at -350. The owner deposits 200, then 500, then withdraws 120. What is the new balance? [Integers]

- **(A)** 330
- **(B)** 130
- **(C)** 350
- **(D)** 1170
- **(E)** -230
- **(F)** 230
- **(G)** -1170

75. A train 200 m long passes a pole in 8 s. What is the train's speed in m/s, and in km/h? [Motion & Time]

- **(A)** 25 km/h only
- **(B)** 2.5 m/s, which is 9 km/h
- **(C)** 25 m/s, which is 90 km/h
- **(D)** 12.5 m/s, which is 45 km/h
- **(E)** 8 m/s, which is 28.8 km/h
- **(F)** 1600 m/s, which is huge
- **(G)** 40 m/s, which is 144 km/h

76. In the expression 6 + 5xy - x, how many terms are there, and what is the coefficient of xy? [Algebraic Expressions]

- **(A)** 2 terms; coefficient 5
- **(B)** 2 terms; coefficient -1
- **(C)** 3 terms; coefficient -1
- **(D)** 3 terms; coefficient 5
- **(E)** 3 terms; coefficient 1
- **(F)** 4 terms; coefficient 5

77. A cyclone ALERT is issued 48 hours before landfall and a WARNING 24 hours before. If landfall is expected at noon on the 12th, when is the alert issued? [Winds, Storms & Cyclones]

- **(A)** Midnight on the 11th
- **(B)** Noon on the 13th
- **(C)** Noon on the 9th
- **(D)** Midnight on the 10th
- **(E)** Noon on the 11th
- **(F)** Noon on the 10th
- **(G)** Noon on the 14th

78. What number must be added to -18 to give -3? [Integers]

- **(A)** 15
- **(B)** -3
- **(C)** -36
- **(D)** 3
- **(E)** -21
- **(F)** 21

79. A boy cycles 12 km in 40 minutes. What is his speed in km/h? [Motion & Time]

- **(A)** 18 km/h
- **(B)** 480 km/h
- **(C)** 20 km/h
- **(D)** 30 km/h
- **(E)** 0.3 km/h
- **(F)** 8 km/h

80. Simplify: 2x squared + 3x - x squared - 5x + 4. [Algebraic Expressions]

(A) x squared - 8x + 4
(B) 3x squared - 2x + 4
(C) x squared + 2x + 4
(D) x squared + 8x + 4
(E) x squared - 2x + 4
(F) -x squared - 2x + 4

81. Which list of moving air is ordered from the GENTLEST to the FIERCEST? [Winds, Storms & Cyclones]

(A) storm, strong wind, breeze, cyclone
(B) storm, cyclone, breeze, strong wind
(C) breeze, strong wind, storm, cyclone
(D) cyclone, breeze, storm, strong wind
(E) cyclone, storm, strong wind, breeze
(F) breeze, cyclone, storm, strong wind
(G) strong wind, breeze, cyclone, storm

82. Evaluate: (-4) x (-5) x (-1) x (-2). [Integers]

(A) -13
(B) -20
(C) 40
(D) 20
(E) 13
(F) -40
(G) 80

83. The minute hand of a clock completes one full round in how much time? [Motion & Time]

(A) 2 hours
(B) 1 minute
(C) 30 minutes
(D) 12 hours
(E) 1 hour
(F) 24 hours
(G) 60 hours

84. A number is x. Write an expression for the sum of this number and the next two consecutive numbers. [Algebraic Expressions]

(A) x + 3
(B) 3x + 3
(C) 3x
(D) 3x + 2
(E) x cubed
(F) 3x + 6
(G) 3x + 1

85. A barometer needle keeps dropping quickly through the afternoon. This usually warns of what? [Winds, Storms & Cyclones]

- **(A)** A solar eclipse
- **(B)** Stronger sunshine
- **(C)** No change at all
- **(D)** An approaching storm
- **(E)** A coming cold but dry day
- **(F)** Only a rise in temperature
- **(G)** A clear, sunny spell

86. A city's record high and low temperatures are 41 deg C and -9 deg C. What is the temperature range (high minus low)? [Integers]

- **(A)** 50 deg C
- **(B)** -50 deg C
- **(C)** 369 deg C
- **(D)** 9 deg C
- **(E)** 41 deg C
- **(F)** 32 deg C
- **(G)** -32 deg C

87. A car's speedometer shows 60 km/h at a particular moment. What does a speedometer measure? [Motion & Time]

- **(A)** The speed at that instant
- **(B)** The time taken so far
- **(C)** The total distance travelled
- **(D)** The fuel used
- **(E)** The average speed of the trip
- **(F)** The acceleration only
- **(G)** The distance per litre

88. Find the value of 5p - 2q when p = -3 and q = 4. [Algebraic Expressions]

- **(A)** 23
- **(B)** -23
- **(C)** 7
- **(D)** -22
- **(E)** 17
- **(F)** -7
- **(G)** -19

89. To reduce cyclone damage, how are coastal homes best built? [Winds, Storms & Cyclones]

- **(A)** Out of light cardboard
- **(B)** Low and strong, firmly anchored, with cyclone shelters nearby
- **(C)** Far inland with no shelters at all
- **(D)** With loose tin sheets for roofs
- **(E)** With heavy unfixed objects on the roof
- **(F)** Tall with large glass windows
- **(G)** Right on the open beach

90. Evaluate: (-90) / [(-3) x (-2) x 5]. [Integers]

- **(A)** -3
- **(B)** 15
- **(C)** -30
- **(D)** 30
- **(E)** -90
- **(F)** -15

91. A bus covers the first 100 km in 2 hours and the next 150 km in 3 hours. What is its average speed? [Motion & Time]

- **(A)** 41.7 km/h
- **(B)** 60 km/h
- **(C)** 5 km/h
- **(D)** 55 km/h
- **(E)** 50 km/h
- **(F)** 45 km/h

92. What is the numerical coefficient of the term -x squared y? [Algebraic Expressions]

- **(A)** 0
- **(B)** x
- **(C)** -2
- **(D)** 1
- **(E)** -1
- **(F)** 2

93. A pendulum-like wind vane on a roof points its arrow towards the WEST. By the rule that a wind is named after the direction it comes FROM, what is this wind called? [Winds, Storms & Cyclones]

- **(A)** an easterly (coming from the east)
- **(B)** a westerly (coming from the west)
- **(C)** a northerly (coming from the north)
- **(D)** a still wind
- **(E)** a downward wind
- **(F)** a westerly coming from the east

94. At midnight the temperature is -4 deg C and it rises a steady 2 deg C every hour. After how many hours will it reach 10 deg C? [Integers]

- **(A)** 7 hours
- **(B)** 28 hours
- **(C)** 3 hours
- **(D)** 5 hours
- **(E)** 14 hours
- **(F)** 12 hours
- **(G)** 6 hours

95. A pendulum completes 20 swings in 50 s. Find its time period, and how many swings it makes in 5 minutes. [Motion & Time]

- **(A)** 0.4 s; 750 swings
- **(B)** 2.5 s; 100 swings
- **(C)** 2.5 s; 20 swings
- **(D)** 2.5 s; 120 swings
- **(E)** 2 s; 150 swings
- **(F)** 5 s; 60 swings

96. Take twice a number m, subtract 7, then double the whole result. Write the expression. [Algebraic Expressions]

- **(A)** 4m - 28
- **(B)** 2m - 7
- **(C)** 4m - 14
- **(D)** 2m - 14
- **(E)** 4m - 7
- **(F)** 4m - 49
- **(G)** 4m + 14

97. Moist air rises from warm ground at 27 deg C and cools by 6 deg C for every 1000 m it climbs. What is its temperature when it reaches a height of 800 m, where cloud begins to form? [Winds, Storms & Cyclones]

- **(A)** 18 deg C
- **(B)** 22.2 deg C
- **(C)** 4.8 deg C
- **(D)** 31.8 deg C
- **(E)** 23.8 deg C
- **(F)** 26.4 deg C
- **(G)** 21.2 deg C

98. A score changes by -1 a total of 99 times in a row, written as a product of 99 factors each equal to -1. What is the result? [Integers]

- **(A)** 99
- **(B)** 1
- **(C)** -2
- **(D)** 0
- **(E)** -99
- **(F)** 2
- **(G)** -1

99. An object moves equal distances in equal intervals of time. What kind of motion is this? [Motion & Time]

- **(A)** Circular motion only
- **(B)** Slowing-down motion
- **(C)** Non-uniform motion
- **(D)** Backward motion
- **(E)** Accelerating motion
- **(F)** Uniform motion (constant speed)

100. Simplify 4ab + 3ab - 2ab, then find its value when a = 2 and b = 5. [Algebraic Expressions]

- **(A)** 5ab; equals 50
- **(B)** 9ab; equals 90
- **(C)** 5ab; equals 20
- **(D)** 5a + b; equals 15
- **(E)** 5ab; equals 100
- **(F)** 5ab; equals 25
- **(G)** ab; equals 10